



**SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)**

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)  
Re-Accredited by NAAC with 'A++' Grade

## ASSIGNMENT No: 3

**TITLE:** Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

### Problem Statement

Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

### Learning Objectives

To implement method overloading and use static variables and methods in Java programs.

### Theory

Method overloading allows multiple methods to have the same name with different parameter lists, enabling compile-time polymorphism. The static keyword is used for variables and methods that belong to the class instead of individual objects. Static members share common data among all objects and reduce memory usage. These concepts improve code flexibility, readability, and efficient resource management.



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## Exercise

1. Develop a Calculator program using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.

## Code

```
experiments - Experiment3/src/Experiment3/Calc.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Calc.java
1 package Experiment3;
2
3
4 public class Calc {
5     // Static variable to count calculations
6     static int calculationCount = 0;
7
8     // Overloaded method for integer addition
9     int add(int a, int b) {
10         calculationCount++;
11         return a + b;
12     }
13
14     // Overloaded method for decimal addition
15     double add(double a, double b) {
16         calculationCount++;
17         return a + b;
18     }
19
20     public static void main(String[] args) {
21         Calc calc = new Calc();
22
23         int sum1 = calc.add(15, 25);
24         double sum2 = calc.add(10.5, 20.7);
25
26         System.out.println("Addition of Integers = " + sum1);
27         System.out.println("Addition of Decimals = " + sum2);
28         System.out.println("Total Calculations = " + calculationCount);
29     }
30 }
31 }
```

## Output

```
experiments - Experiment3/src/Experiment3/Calc.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Problems Javadoc Declaration Console X Sponsor the Eclipse IDE

<terminated> Calc [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.w
Addition of Integers = 40
Addition of Decimals = 31.2
Total Calculations = 2
```



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2. Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

## Code

```
1 package Experiment3;
2 public class Restaurant {
3     // Static variable to count total orders
4     static int totalOrders = 0;
5     // Dine-in bill
6     double calculateBill(double amount) {
7         totalOrders++;
8         return amount;
9     }
10    // Takeaway bill (packing charge)
11    double calculateBill(double amount, double packingCharge) {
12        totalOrders++;
13        return amount + packingCharge;
14    }
15    // Delivery bill (packing + delivery charge)
16    double calculateBill(double amount, double packingCharge, double deliveryCharge) {
17        totalOrders++;
18        return amount + packingCharge + deliveryCharge;
19    }
20    public static void main(String[] args) {
21        Restaurant bill = new Restaurant();
22        System.out.println("Dine-in Bill = ₹" + bill.calculateBill(500));
23        System.out.println("Takeaway Bill = ₹" + bill.calculateBill(500, 20));
24        System.out.println("Delivery Bill = ₹" + bill.calculateBill(500, 20, 40));
25        System.out.println("Total Orders = " + totalOrders);
26    }
27 }
28 }
```

## Output

```
Dine-in Bill = ₹500.0
Takeaway Bill = ₹520.0
Delivery Bill = ₹560.0
Total Orders = 3
```